A conditional bound on the high-frequency tail

Weighted energy + radial-pressure source on the same prescribed long throat

Assumed constant-radius core
and two compact probes

ρc = 32; b = 0.2d
Sharp joins remain unsupported

Frequency bands are schematic, not to scale. x = κℓ; $\hbar$ = c = d = 1.

Infrared

$0 \le x < 10^{-6}$

m = 0 · phase π · absolute bound

Central	**≤ 1.31e-09**
Upper-side	**≤ 2.64e-09**

Finite J=9 / DG; floating enclosure.
Conditional, not interval-verified.
Other angular sectors remain open.

Computed window

$10^{-6} \le x \le 48$

N = 4 · selected phase π

Central	**-0.01151672**
Upper-side	**-0.00442474**

Numerical finite-window integrals.
Mesh and angular limits remain open.

Ultraviolet

$x \ge 48$

All angular modes; both Weyls

Central	**≤ 2.73e-20**
Upper-side	**≤ 1.98e-06**

Interval-evaluated analytic majorant.
Prescribed core + passive return.
Phase-independent absolute bound.

Separate finite-frequency issue: uncomputed angular modes can feed back into the computed modes.

At x = 0.4439, feedback bounds: 118.2 central / 45.46 upper-side (pointwise, per dκ).

The present bounds are too loose to certify the source sign; they are not integrated error bounds.

Bounds and integrals have different qualifications. This ledger does not establish a combined full-source bound.

Absolute supporting stress, a physical junction and stability remain unresolved; no material recipe is established.

No stable medium or 13 Sep 2027 → 12 Sep 2027 passage demonstrated.